GRAND FINALE · HEALTHTECH TRACK

CODEFEST'26 -SYNTECH-

Kawan.

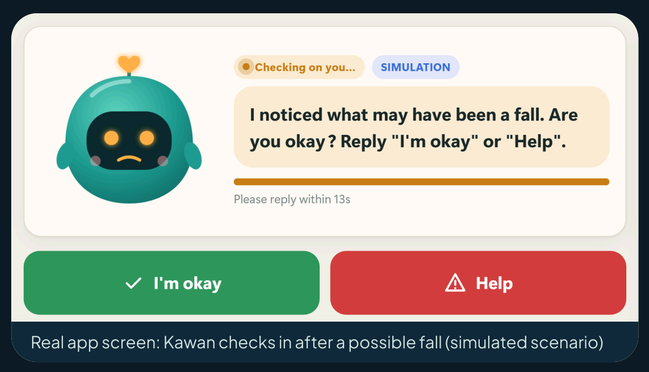
kawan (Malay) · *friend*

An AI guardian for people who live alone: it notices, checks in, and alerts family only when needed.

Not a doctor. Not a surveillance camera. A guardian.

Team Nexora UTP CODEFEST '26

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Real app screen: Kawan checks in after a possible fall (simulated scenario)

01 The problem

IMPACT

Living alone with an illness is risky; help often arrives too late.

15.6%
of Malaysian adults have **diabetes** and **29.2%** hypertension; about 4 in 10 of them don't know it.[1]

684k
people die from **falls** every year. Adults over 60 suffer the most fatal falls.[2]

80%
of fallers over 90 **could not get up alone**; 30% lay on the floor for an hour or more. **Call alarms mostly went unused.**[3]

~50%
of patients with long-term illness **don't take their medicine** as prescribed.[4]

Today's tools wait for the person to **press a button or read a chart**. After a fall, in confusion, or when a dose is forgotten, that is exactly when they can't.

02 Our solution

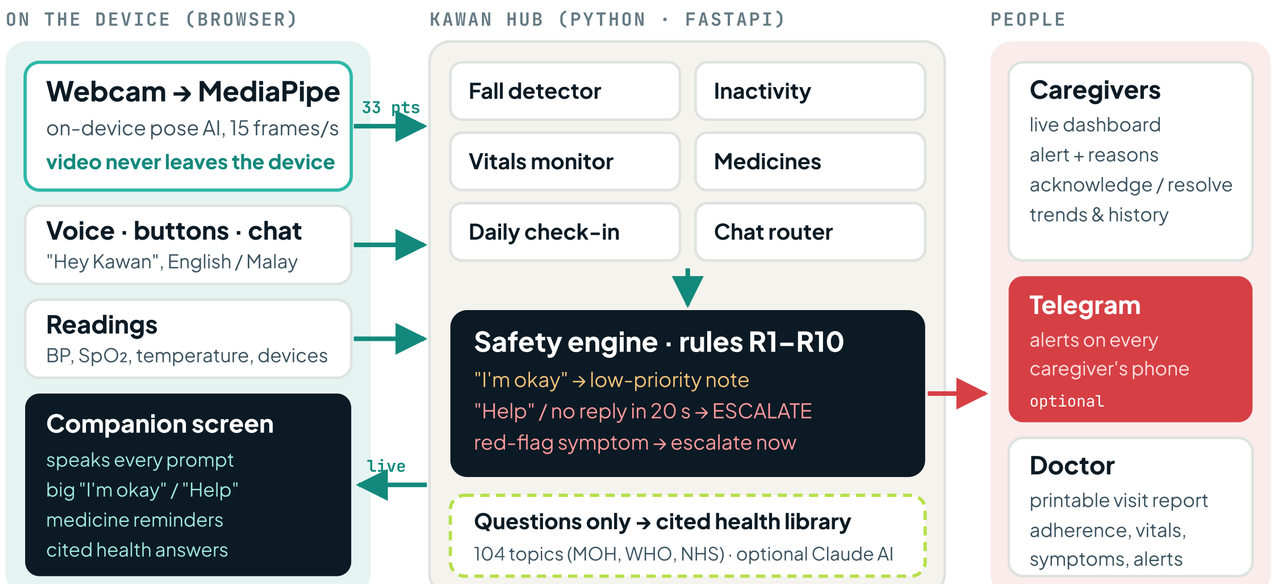
IMPACT · USABILITY

1. **Notice** — The device's own camera spots falls and unusual stillness; medicines and readings are tracked.
2. **Check in** — "**Are you okay?**" by voice, button or chat, in English or Bahasa Melayu.
3. **Escalate only if needed** — "Help", a red-flag symptom, or **no reply in 20 s** alerts every caregiver.
4. **Explain why** — Caregivers see **why** Kawan is worried, and what else it could be.

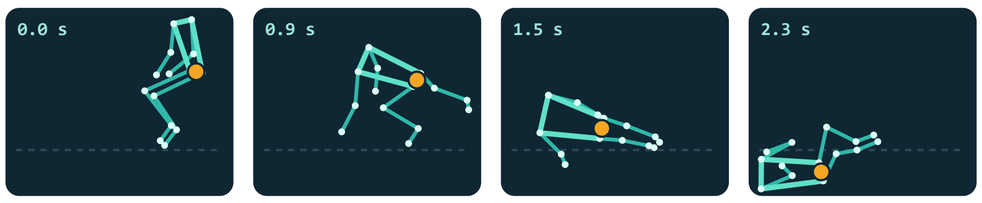
No wearables · No button to press · Works offline · English · Bahasa Melayu

03 How it works

TECHNICAL EXECUTION



Fall detection on a real fall video



upright	tipping over	hips dropping fast	on the floor
torso 11° · hip 0.60	torso 63° · hip 0.64	torso 72° · hip 0.84	torso 87° · hip 1.03

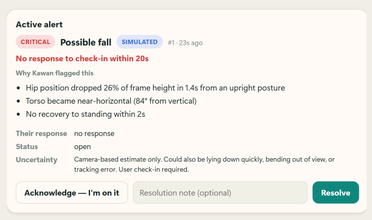
1. **Upright**, then
2. **hips drop fast:** ≥ 0.6 of their own torso length in ≤ 1.5 s (works near or far)
3. **body horizontal** (≥ 60°) or out of view
4. **still down** after 2 s → check in

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Kawan's own output for this clip:
Hip dropped 14% of frame (0.6× torso) in 1.3 s · torso 90° · no recovery within 2 s
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Skeleton extracted from UR Fall clip fall-04.[5]

04 Key features

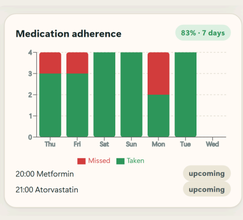
USABILITY · REAL APP SCREENS



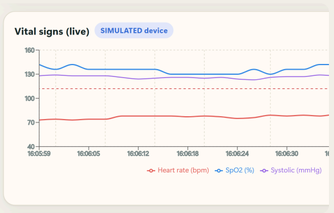
Fall & inactivity detection
Every alert lists its reasons and "what else it could be". Inactivity is judged against this person's normal.



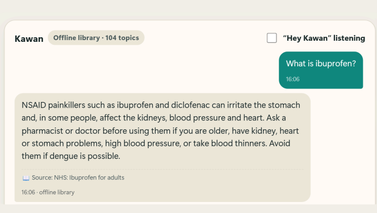
Check in, in English or Malay
Big buttons, spoken prompts, "Hey Kawan" voice. Here on a phone: chest pain reported in Malay → "call 999", caregiver alerted.



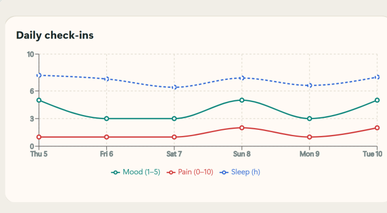
Medicine reminders
Spoken reminders, one tap for "I took it", missed-dose notice to caregivers, adherence chart.



Vitals & daily check-in
Alerts need sustained readings. Check-ins give non-diagnostic triage: self-care → see a doctor → call 999.



Health answers with sources
104 topics from MOH Malaysia guidelines, WHO and NHS. The source is shown under every answer.



Caregiver dashboard & trends
Live status, timeline, mood/pain/sleep trends, alerts to several caregivers' phones, doctor report.

05 Safe by design

- **Rules decide, AI only explains.** The AI never sees a check-in reply.
- **Fails safe.** Unclear replies keep the timer running.
- **Explains itself:** why it fired and what else it could be.
- **Private & offline:** only 33 body points leave the camera, no images are stored, and the core works without internet.
- **Honest data:** nothing invented; demo data is labelled SIMULATED.

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"Okay, but my leg hurts" → counts as HELP
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06 Results

TECHNICAL EXECUTION · USABILITY

FALL DETECTION · REAL VIDEOS[5]

27 / 30 real falls caught (90%)

The earlier frame-size rule caught 17/30. 13 of 40 everyday clips triggered a check-in, 11 of them people lying down on purpose: that's why Kawan **asks first**.

ENGINEERING QUALITY

148 automated tests

117 backend · 19 frontend · 12 browser end-to-end, run by GitHub Actions on every push. Open source (MIT).

EARLY INFORMAL FEEDBACK · 4 TESTERS (3 FRIENDS AND A PARENT)

> *"The reminders are the part I'd use every day."*

> *"It noticed when I fell and asked if I was okay. That's reassuring when no one's home."*

> *"I don't need to read anything. I just talk to it and it talks back."*

> *"I asked what ibuprofen is and how it works, and finally understood it."*

WHAT WE HEARD	WHAT WE DID / WILL DO
2–3 caregivers want alerts on their phones	✓ **Done:** Telegram alerts go to every caregiver
Voice sometimes doesn't catch what's said	Big tap buttons as backup now; better speech recognition next
It knows less than general chatbots	Deliberate: cited sources only, so it can't invent medical facts
Cover emergencies beyond falls	Already alerts on red-flag symptoms and abnormal vitals; more planned

07 Feasibility & next steps

FEASIBILITY

Ready to run today

Hardware: any laptop or tablet with a webcam + Chrome/Edge. Nothing to buy.
Cost: free open-source stack; AI and Telegram optional.
Setup: one command; a secure link opens it on phones.
Maintainable: thresholds in one config file; 148 tests on every change.

Roadmap

NOW — Working prototype with all features above; tried by 4 early users.

NEXT — Caregiver login · hosted multi-household version · push notifications · smartwatch/oximeter · better speech.

LATER — Clinical validation with healthcare partners · more emergency types · low-cost tablet kiosk.

Honest limitations

Detection rules are not clinically validated.
One camera, one person in view.
The browser tab must stay open.
Not a medical device.

REFERENCES

1. Institute for Public Health, MOH Malaysia. *National Health and Morbidity Survey 2023: NCD Fact Sheet*. 2024.
2. World Health Organization. *Falls — Fact sheet*. 2021.
3. Fleming J, Brayne C. Inability to get up after falling… people over 90. *BMJ* 2008;337:a2227.
4. World Health Organization. *Adherence to Long-Term Therapies: Evidence for Action*. 2003.
5. Kwolek B, Kepski M. Human fall detection on embedded platform… *Comput Methods Programs Biomed* 2014;117(3):489–501. UR Fall Detection Dataset, CC BY-NC-SA 4.0.
6. Google. *MediaPipe Pose Landmarker*. ai.google.dev/edge/mediapipe
7. MOH Malaysia Clinical Practice Guidelines (T2DM 2020, Hypertension 2018 and others), as cited in the app's library.

Prototype for UTP CodeFest '26 — not a medical device. In an emergency call 999. Built with the help of Claude Code (AI coding assistant), as disclosed under competition rules.

DEMO VIDEO

SOURCE CODE